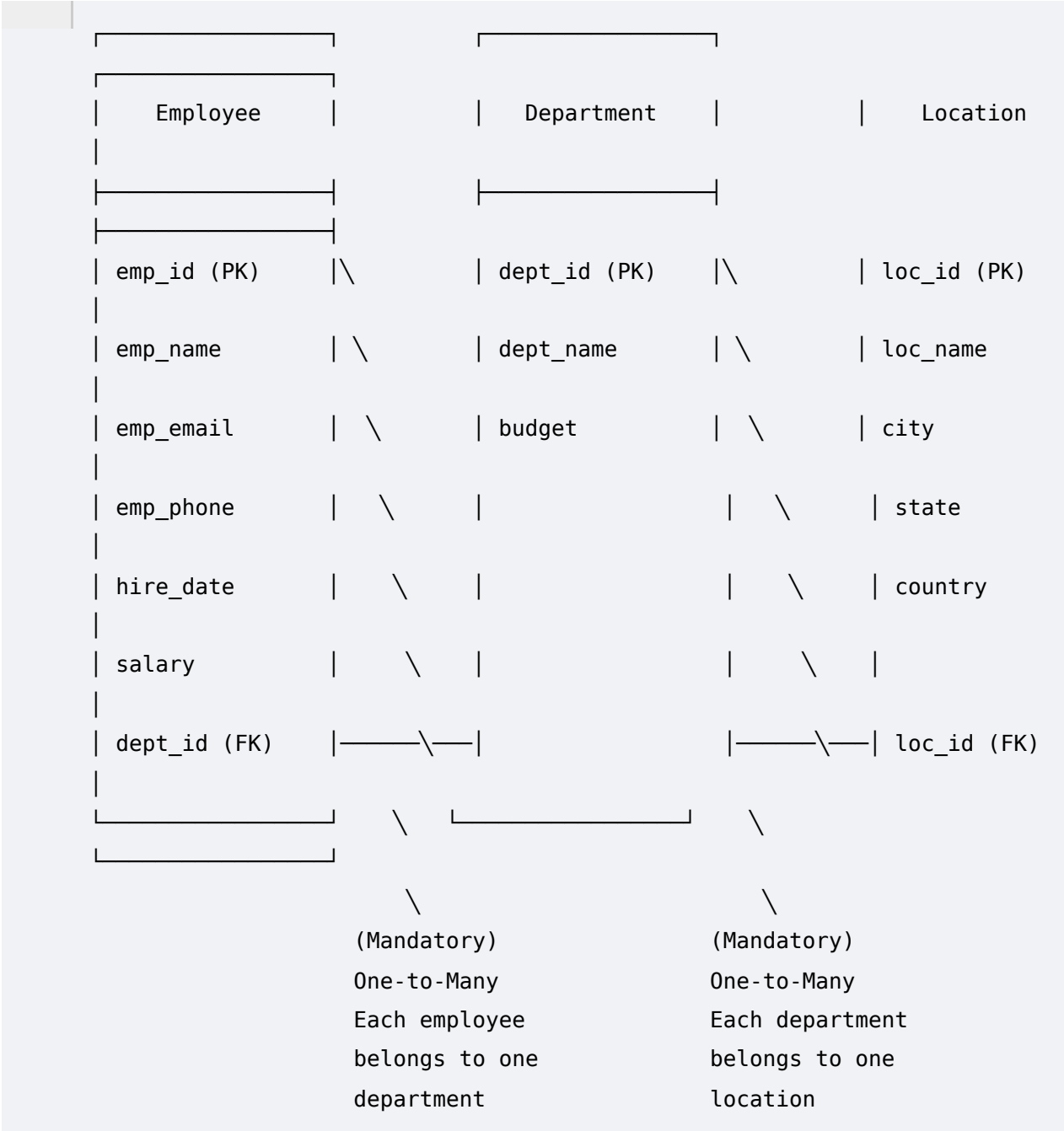


ENTITY-RELATIONSHIP (ER) DIAGRAM

IN CROW'S FOOT NOTATION:

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ALTERNATIVE MERMAID.JS FORMAT (FOR ONLINE VISUALIZATION):

RELATIONSHIP CARDINALITIES

Relationsh ip	Cardinalit y	Explanatio n
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Employee → Department	Many-to-One	Many employees belong to one department
Employee → Location	Many-to-One	Many employees work at one location
Department → Location	Many-to-One	Many departments are in one location

ENTITY ATTRIBUTES

EMPLOYEE ENTITY

Attribute	Datatype	Constraints
emp_id	INT	PRIMARY KEY, AUTO_INCREMENT
emp_name	VARCHAR(100)	NOT NULL

emp_email	VARCHAR(100)	UNIQUE, NOT NULL
emp_phone	VARCHAR(15)	
hire_date	DATE	NOT NULL
salary	DECIMAL (10,2)	
dept_id	INT	FOREIGN KEY (references Department)
loc_id	INT	FOREIGN KEY (references Location)

DEPARTMENT ENTITY

Attribute	Datatype	Constraints
dept_id	INT	PRIMARY KEY, AUTO_INCREMENT

dept_name	VARCHAR(50)	UNIQUE, NOT NULL
budget	DECIMAL (15,2)	
loc_id	INT	FOREIGN KEY (references Location)

LOCATION ENTITY

Attribute	Datatype	Constraints
loc_id	INT	PRIMARY KEY, AUTO_INCREMENT
loc_name	VARCHAR(50)	UNIQUE, NOT NULL
city	VARCHAR(50)	NOT NULL
state	VARCHAR(50)	
country	VARCHAR(50)	NOT NULL

CONVERTED TABLES (SCHEMA)

TABLE 1: LOCATION

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LOCATION	
loc_id (PK)	INT (AUTO_INCREMENT)
loc_name	VARCHAR(50) NOT NULL UNIQUE
city	VARCHAR(50) NOT NULL
state	VARCHAR(50)
country	VARCHAR(50) NOT NULL

TABLE 2: DEPARTMENT

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DEPARTMENT	
dept_id (PK)	INT (AUTO_INCREMENT)
dept_name	VARCHAR(50) NOT NULL UNIQUE
budget	DECIMAL(15,2)
loc_id (FK)	INT REFERENCES Location

TABLE 3: EMPLOYEE

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EMPLOYEE	
emp_id (PK)	INT (AUTO_INCREMENT)
emp_name	VARCHAR(100) NOT NULL
emp_email	VARCHAR(100) UNIQUE NOT NULL
emp_phone	VARCHAR(15)
hire_date	DATE NOT NULL
salary	DECIMAL(10,2)
dept_id (FK)	INT REFERENCES Department
loc_id (FK)	INT REFERENCES Location

SQL CREATE TABLE STATEMENTS

sql

```
-- 1. Create Location table (no foreign keys)
CREATE TABLE Location (
    loc_id INT PRIMARY KEY AUTO_INCREMENT,
    loc_name VARCHAR(50) NOT NULL UNIQUE,
    city VARCHAR(50) NOT NULL,
    state VARCHAR(50),
    country VARCHAR(50) NOT NULL
);

-- 2. Create Department table (references Location)
CREATE TABLE Department (
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_name VARCHAR(50) NOT NULL UNIQUE,
    budget DECIMAL(15,2),
    loc_id INT,
    FOREIGN KEY (loc_id) REFERENCES Location(loc_id)
);

-- 3. Create Employee table (references Department and Location)
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY AUTO_INCREMENT,
    emp_name VARCHAR(100) NOT NULL,
    emp_email VARCHAR(100) UNIQUE NOT NULL,
    emp_phone VARCHAR(15),
    hire_date DATE NOT NULL,
    salary DECIMAL(10,2),
    dept_id INT,
    loc_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
    FOREIGN KEY (loc_id) REFERENCES Location(loc_id)
);
```

SAMPLE DATA (FOR TESTING)

sql

```
-- Insert Locations
INSERT INTO Location (loc_name, city, state, country) VALUES
('HQ', 'New York', 'NY', 'USA'),
```

```

('Branch A', 'Boston', 'MA', 'USA'),
('Branch B', 'London', NULL, 'UK');

-- Insert Departments
INSERT INTO Department (dept_name, budget, loc_id) VALUES
('IT', 1000000.00, 1),
('HR', 500000.00, 1),
('Finance', 750000.00, 2),
('Sales', 600000.00, 3);

-- Insert Employees
INSERT INTO Employee (emp_name, emp_email, emp_phone, hire_date, salary,
dept_id, loc_id) VALUES
('John Doe', 'john@company.com', '555-0101', '2020-01-15', 75000.00, 1,
1),
('Jane Smith', 'jane@company.com', '555-0102', '2019-03-10', 65000.00,
2, 1),
('Bob Johnson', 'bob@company.com', '555-0103', '2021-06-20', 80000.00,
3, 2);

```

KEY RELATIONSHIPS DIAGRAM

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